MIXED VOLUMES OF OKOUNKOV BODIES

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ABSTRACT. Let X be a connected smooth projective variety of dimension n. We prove that the movable product of any big line bundles $L_1, \dots, L_n$ equals $n!$ times the supremum of the mixed volumes of their Okounkov bodies over all valuations.

1. INTRODUCTION

Let X be a connected smooth projective variety of dimension n. For a big line bundle L and a surjective valuation

$$\nu\colon \mathbb{C}(X)^\times \longrightarrow \mathbb{Z}^n_{\mathrm{lex}}$$

that is trivial on $\mathbb{C}^\times$, write $\Delta_\nu(L) \subseteq \mathbb{R}^n$ for the associated Okounkov body. Its Euclidean volume recovers the volume, or equivalently the top movable self-intersection, of L:

$$n! \operatorname{vol} \Delta_\nu(L) = \operatorname{vol}(L) = \langle L^n \rangle. \tag{1.1}$$

Okounkov bodies depend only on numerical classes [LM09, Proposition 4.1]. Conversely, Jow proved that two big divisors on a normal complex projective variety are numerically equivalent if their Okounkov bodies agree for every admissible flag [Jow10, Theorem A]. Thus the collection of all Okounkov bodies is a faithful numerical invariant.

The first conclusion of this paper is that this collection also recovers mixed movable products.

Theorem 1.1. *Let $L_1, \dots, L_n$ be big line bundles on X, then*

$$\langle L_1, \dots, L_n \rangle = n! \sup_\nu \operatorname{vol}\Big(\Delta_\nu(L_1), \dots, \Delta_\nu(L_n)\Big), \tag{1.2}$$

where ν ranges over the valuations above.

Here vol denotes mixed volume, normalized by $\operatorname{vol}(K, \dots, K) = \operatorname{vol}(K)$. For ample line bundles, Wilms proved that a suitable single admissible flag realizes the mixed intersection number [Wil25, Theorem 1.1].

We now pass to its singular-metric refinement. A Hermitian big line bundle (L, h) will mean a big line bundle endowed with a singular psh metric with positive volume (in the sense of Cao). In [Xia25], the author introduced the partial Okounkov body $\Delta_\nu(L, h) \subseteq \mathbb{R}^n$ extending the usual Okounkov body construction. The volume theorem [Xia25, Theorem A] gives

$$n! \operatorname{vol} \Delta_\nu(L, h) = \operatorname{vol} \mathrm{dd^c} h, \tag{1.3}$$

where the volume on the right-hand side is taken in the sense of Cao [Cao14]. Moreover, the family of partial Okounkov bodies over all rank-n valuations determines the $\mathcal{I}$-singularity type of h [Xia25, Theorem B], a metric analogue of Jow's theorem.

A formula of this type was first proposed in [Xia25, Conjecture 8.8]. We prove it here.

Theorem 1.2. *Let X be a connected smooth projective variety of dimension n. For $1 \le i \le n$, let L_i be a big line bundle and let h_i be a psh metric on L_i whose volume is positive. Then*

$$\operatorname{vol}\left(\mathrm{dd^c} h_1, \dots, \mathrm{dd^c} h_n\right) = n! \sup_\nu \operatorname{vol}\Big(\Delta_\nu(L_1, h_1), \dots, \Delta_\nu(L_n, h_n)\Big). \tag{1.4}$$

Here ν runs over all valuations as above.

Again, the mixed volume is taken in the sense of Cao, as shown in [Xia26b], it coincides with the mixed volume in the sense of Darvas–Xia as recalled in Section 2. Taking each h_i to have minimal singularities reduces Theorem 1.2 to Theorem 1.1, while its diagonal specialization is (1.3).

The proof of the $\leq$ direction in (1.4) can be easily reduced to Wilms' theorem [Wil25, Theorem 1.1]. Conversely, the $\geq$ direction can be reduced to the $\geq$ direction in (1.2). We shall apply a suitable toric degeneration to further reduce to the toric case, where the results are essentially known.

Acknowledgments. The author used the Rethlas agent, backed by an OpenAI GPT-5.6 Sol model, for proof exploration and consistency checks during the development of this work. AI-generated suggestions were treated as provisional; the author assumes full responsibility for every argument and for the final text.

The author is supported by the National Key R&D Program of China under Grant No. 2025YFA1018200.

2. Preliminaries

We follow the language and notation of [Xia26a]. All products of closed positive currents are non-pluripolar products. We use $\mathrm{dd^c} = (2\pi)^{-1}\mathrm{i}\partial\bar{\partial}$, and $\mathbb{N} = \{0, 1, 2, \dots\}$. The notation $\mathbb{Z}^n_{\mathrm{lex}}$ means $\mathbb{Z}^n$ with the lexicographic order, and $e_1, \dots, e_n$ denotes its standard basis. A superscript $^\times$ means that zero is omitted, and $\operatorname{Conv} A$ denotes the convex hull of a subset $A \subseteq \mathbb{R}^n$. We use $\mathbb{P}(V)$ for the space of lines in a vector space V.

2.1. **Singularities and partial Okounkov bodies.** Fix a smooth Hermitian metric h^0 on a big line bundle L and write

$$\theta \coloneqq c_1(L, h^0), \qquad h = h^0 \mathrm{e}^{-\varphi}, \qquad \theta_\varphi \coloneqq \theta + \mathrm{dd^c}\varphi,$$

where $\mathrm{PSH}(X, \theta)$ is the set θ-psh functions, and

$$\mathrm{PSH}(X, \theta)_{>0} \coloneqq \left\{ u \in \mathrm{PSH}(X, \theta) : \int_X \theta_u^n > 0 \right\}.$$

For $u, v \in \mathrm{PSH}(X, \theta)$, write $u \preceq v$ if $u \leq v + O(1)$. The model envelope is

$$P_\theta[u] \coloneqq \sup{}^* \{ v \in \mathrm{PSH}(X, \theta) : v \leq 0,\ v \preceq u \},$$

where $\sup^*$ denotes the upper-semicontinuous regularization of the pointwise supremum. A positive-mass potential u is a model potential if $P_\theta[u] = u$. Two positive-mass potentials are P-equivalent if their model envelopes agree. They are $\mathcal{I}$-equivalent if $\mathcal{I}(\lambda u) = \mathcal{I}(\lambda v)$ for every $\lambda > 0$, where $\mathcal{I}(\lambda u)$ is the multiplier ideal sheaf. The $\mathcal{I}$-model envelope is

$$P_\theta[u]_{\mathcal{I}} \coloneqq \sup{}^* \{ v \in \mathrm{PSH}(X, \theta) : v \leq 0,\ v \text{ is } \mathcal{I}\text{-equivalent to } u \}.$$

A potential $\varphi \in \mathrm{PSH}(X, \theta)_{>0}$ is called $\mathcal{I}$-good if

$$P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}}.$$

The corresponding volume is

$$\operatorname{vol} \theta_\varphi \coloneqq \int_X \theta^n_{P_\theta[\varphi]_{\mathcal{I}}}.$$

More generally, if $L_1, \dots, L_n$ are big line bundles with θ_i being a closed smooth real $(1,1)$-form in $c_1(L_i)$ and $\varphi_i \in \mathrm{PSH}(X, \theta_i)_{>0}$, we write

$$\operatorname{vol}\left(\theta_1 + \mathrm{dd^c}\varphi_1, \dots, \theta_n + \mathrm{dd^c}\varphi_n\right) = \int_X \left(\theta_1 + \mathrm{dd^c} P_{\theta_1}\left[\varphi_1\right]_{\mathcal{I}}\right) \wedge \cdots \wedge \left(\theta_n + \mathrm{dd^c} P_{\theta_n}\left[\varphi_n\right]_{\mathcal{I}}\right).$$

The mixed volume defined here coincides with that defined by Cao [Cao14].

By [Xia26a, Theorem 7.1.1], $\mathcal{I}$-goodness is equivalent both to the mass–volume identity

$$\int_X \theta_\varphi^n = \operatorname{vol} \theta_\varphi$$

and to the existence of potentials φ_j with analytic singularities such that

$$\varphi_j \xrightarrow{d_S} \varphi.$$

Here d_S is the pseudometric on singularity types introduced in [DDNL21], see also [Xia26a, Definition 6.2.1]. The approximants may be chosen so that every θ_{φ_j} is a Kähler current.

Following [Xia26a, Definition 1.6.1], a potential has analytic singularities if locally it is of the form

$$c\log(|f_1|^2+\cdots+|f_N|^2)+O(1),\qquad c\in\mathbb{Q}_{>0},$$

where the f_j are holomorphic and $O(1)$ is locally bounded.

Let $\nu\colon \mathbb{C}(X)^\times\to\mathbb{Z}^n_{\mathrm{lex}}$ be surjective and trivial on $\mathbb{C}^\times$. Its center exists on X by properness. Choose a rational section τ of L which is regular and nowhere vanishing near the center. For a section s of L^k, set $\nu_\tau(s)\coloneqq\nu(s/\tau^k)$, and suppress the subscript τ below. For every $k\geq 1$ for which the space below is non-zero, set

$$\Delta_{\nu,k}(\theta,\varphi)\coloneqq\operatorname{Conv}\left\{k^{-1}\nu(s) : s\in \mathrm{H}^0(X,L^k\otimes\mathcal{I}(k\varphi))^\times\right\}. \tag{2.1}$$

Let $\Gamma(\theta,\varphi)\subseteq\mathbb{Z}^n\times\mathbb{N}$ consist of $(0,0)$ and all pairs $(\nu(s),k)$ occurring above. By [Xia26a, Theorem 10.3.1], $\Gamma(\theta,\varphi)$ is an almost semigroup. Using the continuous Okounkov-body map Δ of [Xia26a, Theorem 10.1.2], define $\Delta_\nu(\theta,\varphi)=\Delta_\nu(L,h)\coloneqq\Delta(\Gamma(\theta,\varphi))$. Changing the rational trivialization translates all the bodies in (2.1); none of the mixed volumes below is affected.

We record the properties used in the proof. We write $\varphi\preceq_{\mathcal{I}}\psi$ when $\mathcal{I}(\lambda\varphi)\subseteq\mathcal{I}(\lambda\psi)$ for every $\lambda>0$. Let $\mathcal{K}_n$ be the space of non-empty compact convex subsets of $\mathbb{R}^n$, and let d_{Haus} be the Hausdorff distance on $\mathcal{K}_n$. We write $\operatorname{vol}(K)$ for the Euclidean n-volume of $K\in\mathcal{K}_n$. First,

$$\operatorname{vol}\Delta_\nu(\theta,\varphi)=\frac{1}{n!}\operatorname{vol}\theta_\varphi \tag{2.2}$$

by [Xia26a, Corollary 10.3.2]. The right-hand side is independent of ν. Second, the map

$$\Delta_\nu(\theta,\cdot)\colon(\mathrm{PSH}(X,\theta)_{>0},d_S)\longrightarrow(\mathcal{K}_n,d_{\mathrm{Haus}})$$

is continuous by [Xia26a, Theorem 10.3.2]. Third, for another Hermitian big line bundle (L',h'), singularity comparison and tensor products give

$$\varphi\preceq_{\mathcal{I}}\psi\Longrightarrow\Delta_\nu(\theta,\varphi)\subseteq\Delta_\nu(\theta,\psi), \tag{2.3}$$

$$\Delta_\nu(L,h)+\Delta_\nu(L',h')\subseteq\Delta_\nu(L\otimes L',h\otimes h'), \tag{2.4}$$

by [Xia26a, Propositions 10.3.8 and 10.3.10]. In particular, the partial body is contained in the ordinary Okounkov body $\Delta_\nu(L)$, obtained from the same construction using the complete linear series $\mathrm{H}^0(X,L^k)$. The latter satisfies

$$\operatorname{vol}\Delta_\nu(L)=\frac{1}{n!}\operatorname{vol}(L) \tag{2.5}$$

by [Xia26a, Corollary 10.3.1], where $\operatorname{vol}(L)\coloneqq\lim_{k\to\infty}n!k^{-n}h^0(X,L^k)$.

We will also use basic facts about model envelopes. By [Xia26a, Corollary 3.1.2], $P_\theta[\varphi]$ is a model potential. The equality $d_S(\varphi,P_\theta[\varphi])=0$ follows from [Xia26a, Proposition 6.2.2], and P-equivalent potentials are $\mathcal{I}$-equivalent by [Xia26a, Proposition 3.2.9]. Consequently, replacing a potential by its model envelope changes neither its d_S-class nor its partial Okounkov body. If φ has analytic singularities, then so does $P_\theta[\varphi]$ by [Xia26a, Proposition 3.2.10].

Lemma 2.1. *Let $u_j,u\in\mathrm{PSH}(X,\theta)_{>0}$ be model potentials such that $u_j\xrightarrow{d_S}u$. The sequence $(u_j)_j$ has a further subsequence, still indexed by j, for which there are increasing model potentials η_j and decreasing model potentials ψ_j satisfying*

$$\eta_j\leq u_j\leq\psi_j,\qquad \eta_j\leq u\leq\psi_j,\qquad \eta_j\xrightarrow{d_S}u,\qquad \psi_j\xrightarrow{d_S}u.$$

Proof. By [Xia26a, Corollary 6.2.5], the masses $\int_X\theta_{u_j}^n$ have a positive lower bound after finitely many terms. So we can apply [Xia26a, Proposition 6.2.3] (and then take the P-envelope) to obtain η_j and ψ_j. Only the inequality $\eta_j\leq u\leq\psi_j$ needs an extra proof. It follows from $\eta_j\leq u_j\leq\psi_j$ and [Xia26a, Proposition 6.2.5]. $\square$

2.2. **Mixed volumes and movable intersections.** We retain the mixed-volume normalization fixed in Theorem 1.2. Mixed volume is symmetric, multilinear for Minkowski addition, monotone, translation invariant, and continuous for the Hausdorff topology; see [Sch14, Section 5.1].

For big classes $\alpha_1, \dots, \alpha_n \in \mathrm{H}^{1,1}(X, \mathbb{R})$, we write $\langle \alpha_1, \dots, \alpha_n \rangle$ for their movable intersection number [BDPP13]. If $\mu\colon Y \to X$ is a modification with Y smooth and

$$\mu^*\alpha_i = \beta_i + \{E_i\}, \qquad \beta_i \text{ nef}, \quad E_i \text{ is an effective } \mathbb{R}\text{-divisor},$$

then

$$\beta_1 \cdots \beta_n \leq \langle \alpha_1, \dots, \alpha_n \rangle. \tag{2.6}$$

For nef α_i, the movable and ordinary intersection numbers agree.

3. An algebraic upper bound

Lemma 3.1. *Let $F/\mathbb{C}$ be a finitely generated field of transcendence degree n, let X be an integral projective model with $\mathbb{C}(X) = F$, and let ν be a surjective valuation which is trivial on $\mathbb{C}^\times$, written as*

$$\nu\colon F^\times \longrightarrow \mathbb{Z}^n_{\mathrm{lex}}.$$

Let $W_1, \dots, W_n \subset F$ be non-zero finite-dimensional vector spaces. Evaluation of rational functions defines

$$\Phi_i\colon X \dashrightarrow \mathbb{P}(W_i^\vee), \qquad x \longmapsto [\mathrm{ev}_{i,x}], \qquad \mathrm{ev}_{i,x}(f) := f(x).$$

This map is defined wherever all elements of W_i are regular and $\mathrm{ev}_{i,x} \neq 0$. Put

$$\Phi := (\Phi_1, \dots, \Phi_n)\colon X \dashrightarrow \prod_{i=1}^n \mathbb{P}(W_i^\vee),$$

and let Y be the Zariski closure of its image. Assume that Φ is birational onto Y and that

$$\Lambda := \mathrm{span}_{\mathbb{Z}}\{\nu(f) - \nu(g) : f, g \in W_i^\times,\ 1 \leq i \leq n\} = \mathbb{Z}^n.$$

Put $P_i := \mathrm{Conv}\,\nu(W_i^\times)$. If π_i is the projection of the ambient product onto $\mathbb{P}(W_i^\vee)$, put

$$H_i := c_1\big(\pi_i^*\mathcal{O}_{\mathbb{P}(W_i^\vee)}(1)\big).$$

Then

$$n!\,\mathrm{vol}(P_1, \dots, P_n) \leq (H_1 \cdots H_n \cdot Y). \tag{3.1}$$

Proof. For $a \in \mathbb{Z}^n$, let

$$\mathcal{F}_{\succeq a} := \{0\} \cup \{f \in F^\times : \nu(f) \succeq a\}, \qquad \mathcal{F}_{\succ a} := \{0\} \cup \{f \in F^\times : \nu(f) \succ a\},$$

where $\succeq$ is the lexicographic order. The associated graded ring and the initial form of $f \in F^\times$ are

$$\mathrm{gr}_\nu F := \bigoplus_{a \in \mathbb{Z}^n} \mathcal{F}_{\succeq a}/\mathcal{F}_{\succ a}, \qquad \mathrm{in}_\nu(f) := f \bmod \mathcal{F}_{\succ \nu(f)}.$$

The quotient $\mathcal{F}_{\succeq a}/\mathcal{F}_{\succ a}$ is called the leaf of ν at a. Since ν has rational rank n, Abhyankar's inequality forces its residue field to be $\mathbb{C}$; hence every non-zero graded piece is one-dimensional over $\mathbb{C}$. Equivalently, every non-zero finite-dimensional subspace $V \subset F$ satisfies

$$\#\nu(V^\times) = \dim_{\mathbb{C}} V;$$

compare [LM09, Lemma 1.3] and [Xia26a, Section 10.3].

Put $r_i := \dim_{\mathbb{C}} W_i - 1$. Choose one $f_{ij} \in W_i^\times$ above each value in $\nu(W_i^\times)$ and set

$$a_{ij} := \nu(f_{ij}), \qquad 0 \leq j \leq r_i.$$

The a_{ij} are pairwise distinct, so the valuation axiom makes the f_{ij} linearly independent. They therefore form a basis of W_i, and

$$\nu(W_i^\times) = \{a_{i0}, \dots, a_{ir_i}\}.$$

The initial degeneration. Choose $u_\ell \in F^\times$ with $\nu(u_\ell) = e_\ell$. One-dimensionality of the leaves gives a non-canonical graded isomorphism. More precisely, for $a = (a_1, \dots, a_n) \in \mathbb{Z}^n$, write $u^a \coloneqq u_1^{a_1} \cdots u_n^{a_n}$, and write

$$\mathbb{C}[\mathbb{Z}^n] \coloneqq \bigoplus_{a \in \mathbb{Z}^n} \mathbb{C}\chi^a, \qquad \chi^a \chi^{a'} = \chi^{a+a'},$$

for the group algebra of $\mathbb{Z}^n$. Then

$$\mathbb{C}[\mathbb{Z}^n] \xrightarrow{\sim} \operatorname{gr}_\nu F, \qquad \chi^a \longmapsto \operatorname{in}_\nu(u^a).$$

After rescaling the f_{ij}, which merely changes the homogeneous coordinates, we may and do assume that

$$\operatorname{in}_\nu(f_{ij}) = \chi^{a_{ij}}.$$

Let x_{ij} be the homogeneous coordinate on $\mathbb{P}(W_i^\vee)$ dual to f_{ij}. In these coordinates,

$$\Phi_i(x) = [f_{i0}(x) : \cdots : f_{ir_i}(x)].$$

Let $\epsilon_1, \dots, \epsilon_n$ be the standard basis of the multidegree lattice $\mathbb{Z}^n$. The multihomogeneous coordinate ring of the ambient product is

$$S \coloneqq \mathbb{C}[x_{ij} : 1 \le i \le n,\ 0 \le j \le r_i], \qquad \deg x_{ij} \coloneqq \epsilon_i.$$

Let $I \subseteq S$ be the multihomogeneous prime ideal of Y. Equivalently, a multihomogeneous polynomial G belongs to I exactly when $G(f_{ij}) = 0$ in F.

Recall that each a_{ij} is a column vector in $\mathbb{Z}^n$. Put $N \coloneqq \sum_{i=1}^n (r_i + 1)$, and define A to be the $n \times N$ matrix whose column indexed by (i, j) is a_{ij}. Equivalently,

$$\mathsf{A} \coloneqq \left[a_{10} \ \cdots \ a_{1r_1} \mid \cdots \mid a_{n0} \ \cdots \ a_{nr_n}\right] \in \mathbb{Z}^{n \times N}.$$

For $\alpha = (\alpha_{ij}) \in \mathbb{N}^N$, write $x^\alpha \coloneqq \prod_{i,j} x_{ij}^{\alpha_{ij}}$, and define

$$\operatorname{wt}_{\mathsf{A}}(x^\alpha) \coloneqq \sum_{i,j} \alpha_{ij} a_{ij} \in \mathbb{Z}^n.$$

These weights are compared lexicographically. If $0 \neq G = \sum_\alpha \gamma_\alpha x^\alpha$, let $\operatorname{in}_{\mathsf{A}}(G)$ be the sum of the terms of least A-weight, and put

$$Q \coloneqq \operatorname{in}_{\mathsf{A}}(I) \coloneqq \big(\operatorname{in}_{\mathsf{A}}(G) : 0 \neq G \in I\big).$$

Let $t_1, \dots, t_n$ be algebraically independent variables of multidegrees $\deg t_i = \epsilon_i$, and define

$$J \coloneqq \ker\left(S \longrightarrow \mathbb{C}[\mathbb{Z}^n][t_1, \dots, t_n], \quad x_{ij} \longmapsto \chi^{a_{ij}} t_i\right).$$

If $G \in I$ is multihomogeneous, then $G(f_{ij}) = 0$. Taking initial forms of the terms of least valuation in this equality gives $\operatorname{in}_{\mathsf{A}}(G) \in J$. Since I is multihomogeneous, the initial forms of its multihomogeneous elements generate Q; hence

$$Q \subseteq J.$$

This is the argument of [KM19, Lemma 2.16].

By [KM19, Proposition 8.10], there is a rational weight vector $\omega = (\omega_{ij})$ such that $Q = \operatorname{in}_\omega(I)$, where in_ω is defined by taking terms of least ω-weight. After clearing denominators, the ω_{ij} may be taken integral. The weight filtration on S/I respects the $\mathbb{N}^n$-grading and satisfies

$$\operatorname{gr}_\omega(S/I) \simeq S/\operatorname{in}_\omega(I) = S/Q.$$

Each multihomogeneous piece is finite-dimensional, so passing to its associated graded preserves dimension. Hence

$$\dim_{\mathbb{C}}(S/I)_{\mathbf{d}} = \dim_{\mathbb{C}}(S/Q)_{\mathbf{d}} \qquad (\mathbf{d} \in \mathbb{N}^n). \tag{3.2}$$

The toric component. Let $T \coloneqq (\mathbb{C}^\times)^n$. For $z = (z_1, \dots, z_n) \in T$ and $a = (a_1, \dots, a_n) \in \mathbb{Z}^n$, write $z^a \coloneqq z_1^{a_1} \cdots z_n^{a_n}$, and define

$$\tau \colon T \longrightarrow \prod_{i=1}^n \mathbb{P}(W_i^\vee), \qquad z \longmapsto \left([z^{a_{i0}} : \cdots : z^{a_{ir_i}}]\right)_{i=1}^n.$$

If $G \in S$ is multihomogeneous of multidegree $\mathbf{d} = (d_1, \ldots, d_n)$, then its image under the homomorphism defining J is

$$t_1^{d_1} \cdots t_n^{d_n} G(\chi^{a_{ij}}).$$

Thus J is precisely the multihomogeneous ideal of the closure of $\tau(T)$, and

$$Z \coloneqq \operatorname{MultiProj}(S/J) = \overline{\tau(T)}.$$

On the dense torus of the ambient product, τ is defined by the characters $z^{a_{ij}-a_{i0}}$. Their exponents generate $\Lambda = \mathbb{Z}^n$, so the induced homomorphism of character rings is surjective. Hence τ is a closed immersion into that torus. We identify T with its image, a dense open subset of Z; in particular, Z is irreducible of dimension n.

Put $Y_0 \coloneqq \operatorname{MultiProj}(S/Q)$. The inclusion $Q \subseteq J$ gives a closed immersion $Z \hookrightarrow Y_0$. By the multigraded Hilbert theorem, (3.2) gives a common multigraded Hilbert polynomial for Y and Y_0. Its restriction to the diagonal is the Hilbert polynomial of the ample line bundle $\mathcal{O}(1, \ldots, 1)$; hence $\dim Y_0 = \dim Y = n$. Consequently, Z is a top-dimensional irreducible component of Y_0.

For an n-dimensional closed subscheme V of the ambient product, the part of total degree n of its multigraded Hilbert polynomial is

$$\frac{1}{n!}\left(\left(\sum_{i=1}^n d_i H_i\right)^n \cdot [V]_n\right),$$

where $[V]_n$ is its n-dimensional fundamental cycle. Comparing the coefficient of $d_1 \cdots d_n$ yields

$$(H_1 \cdots H_n \cdot Y) = (H_1 \cdots H_n \cdot [Y_0]_n).$$

Write

$$[Y_0]_n = \sum_W m_W [W], \qquad m_W \coloneqq \ell(\mathcal{O}_{Y_0, \eta_W}) \geq 1.$$

Here W runs over the n-dimensional irreducible components of Y_0 and η_W is the generic point of W. In particular, Z occurs with coefficient $m_Z \geq 1$. Since the H_i are globally generated, their mixed degree on every component is non-negative. Therefore

$$(H_1 \cdots H_n \cdot Y) \geq m_Z (H_1 \cdots H_n \cdot Z) \geq (H_1 \cdots H_n \cdot Z). \tag{3.3}$$

The mixed degree of the toric component. The i-th exponent polytope of τ is P_i. General hyperplanes pull back to general Laurent polynomials with Newton polytopes $P_1, \ldots, P_n$. Since their systems are basepoint-free and $\dim(Z \setminus T) \leq n - 1$, an incidence count shows that their common zero set avoids the boundary. The Bernstein–Kushnirenko theorem, or equivalently the toric mixed-intersection formula [Ber75; Ful93], gives

$$(H_1 \cdots H_n \cdot Z) = n! \operatorname{vol}(P_1, \ldots, P_n).$$

Together with (3.3), this proves (3.1). □

Proposition 3.2. *Let X be a smooth connected projective variety of dimension n, let $D_1, \ldots, D_n$ be big $\mathbb{Q}$-divisors, and let $\nu \colon \mathbb{C}(X)^\times \to \mathbb{Z}^n_{\text{lex}}$ be surjective and trivial on $\mathbb{C}^\times$. Then*

$$n! \operatorname{vol}\Big(\Delta_\nu(D_1), \ldots, \Delta_\nu(D_n)\Big) \leq \langle D_1, \ldots, D_n \rangle.$$

Here $\Delta_\nu(D)$ for a big $\mathbb{Q}$-divisor is defined by homogeneity after clearing denominators.

Proof. By homogeneity, assume that all D_i are integral. Let x_ν be the center of ν on X. For each i, choose a non-zero rational section τ_i of $\mathcal{O}_X(D_i)$ which is regular and nowhere vanishing near x_ν. For every integer $k \geq 1$, put

$$W_{i,k} \coloneqq \mathrm{H}^0(X, \mathcal{O}_X(kD_i))\tau_i^{-k} \subset \mathbb{C}(X).$$

Whenever $W_{i,k} \neq 0$, put

$$P_{i,k} \coloneqq \operatorname{Conv} \nu(W_{i,k}^\times).$$

Let Φ_k be the joint evaluation map associated with the $W_{i,k}$, as in Lemma 3.1:

$$\Phi_k = (\Phi_{1,k}, \ldots, \Phi_{n,k}) \colon X \dashrightarrow \prod_{i=1}^n \mathbb{P}(W_{i,k}^\vee).$$

Under multiplication by τ_i^k, its i-th component is the Kodaira map of $|kD_i|$.

The proof of [Xia26a, Proposition 10.3.1] supplies an integer $p_0 > 0$ and $s_0, s_1, \dots, s_n \in W_{1,p_0}^\times$ with

$$\nu(s_j) - \nu(s_0) = e_j, \qquad 1 \le j \le n.$$

Choose a sufficiently divisible multiple $p = rp_0$ such that every map defined by $|pD_i|$ is birational. Set

$$\widetilde{s}_0 := s_0^r, \qquad \widetilde{s}_j := s_j s_0^{r-1} \quad (1 \le j \le n).$$

These elements lie in $W_{1,p}^\times$ and satisfy $\nu(\widetilde{s}_j) - \nu(\widetilde{s}_0) = e_j$. If $k = mp$, then $\widetilde{s}_0^m, \widetilde{s}_j\widetilde{s}_0^{m-1} \in W_{1,k}^\times$, so

$$\operatorname{span}_{\mathbb{Z}}\{\nu(f) - \nu(g) : f, g \in W_{i,k}^\times,\ 1 \le i \le n\} = \mathbb{Z}^n.$$

Moreover, the image of the multiplication map

$$\operatorname{Sym}^m W_{i,p} \longrightarrow W_{i,k}$$

is a linear subsystem defining the m-th Veronese of the birational map associated with $|pD_i|$. Hence $\Phi_{i,k}$, and therefore Φ_k, is birational onto its image.

Let Y_k be the closure of the image of Φ_k, and let $H_{i,k}$ be the restriction to Y_k of the hyperplane class from the i-th factor. Let $\mu_k \colon X_k \to X$ be a common resolution of the base ideals of the systems $|kD_i|$, with X_k smooth, and write

$$\mu_k^*(kD_i) = M_{i,k} + F_{i,k}, \qquad F_{i,k} \ge 0,$$

where $M_{i,k}$ is basepoint-free and $F_{i,k}$ is the effective fixed part. The induced morphism $X_k \to Y_k$ is birational and pulls $H_{i,k}$ back to $M_{i,k}$. Hence the projection formula and Lemma 3.1 give

$$\begin{aligned} n! \operatorname{vol}(P_{1,k}, \dots, P_{n,k}) &\le (H_{1,k} \cdots H_{n,k} \cdot Y_k) \\ &= M_{1,k} \cdots M_{n,k} \\ &\le k^n \langle D_1, \dots, D_n \rangle. \end{aligned}$$

The last inequality is (2.6), since $\mu_k^* D_i - k^{-1} M_{i,k} = k^{-1} F_{i,k} \ge 0$.

Finally, [Xia26a, Proposition 10.3.4] gives

$$k^{-1} P_{i,k} = \Delta_{\nu,k}(D_i) \xrightarrow{d_{\text{Haus}}} \Delta_\nu(D_i).$$

This holds in particular along $k \in p\mathbb{Z}_{>0}$. Divide the preceding inequality by k^n and use continuity of mixed volume. □

4. Uniformity in the valuation

We first isolate a convex-geometric compactness statement. It is important that neither the position nor the shape of the bodies is fixed. Recall that $\mathcal{K}_n$ is the set of convex bodies in $\mathbb{R}^n$.

Lemma 4.1. *For $j \in \mathbb{Z}_{>0}$ and $1 \le i \le n$, let $A_{i,j}, C_{i,j}, D_{i,j}, B_{i,j} \in \mathcal{K}_n$ satisfy*

$$A_{i,j} \subseteq C_{i,j}, D_{i,j} \subseteq B_{i,j}.$$

Suppose that there are constants $v, M > 0$, independent of i and j, such that

$$\begin{aligned} \operatorname{vol}(A_{i,j}) &\ge v, \\ \operatorname{vol}(B_{1,j} + \cdots + B_{n,j}) &\le M, \\ \operatorname{vol}(B_{i,j}) - \operatorname{vol}(A_{i,j}) &\longrightarrow 0 \quad (1 \le i \le n). \end{aligned}$$

Then

$$|\operatorname{vol}(C_{1,j}, \dots, C_{n,j}) - \operatorname{vol}(D_{1,j}, \dots, D_{n,j})| \longrightarrow 0.$$

We refer to [Sch14] for John's theorem and Blaschke selection theorem used in the proof below.

Proof. Translate each i-th quadruple by the same vector so that $0 \in A_{i,j}$, and set $S_j := B_{1,j} + \cdots + B_{n,j}$. These translations do not change mixed volumes. Every body is now contained in S_j, and

$$v \leq \operatorname{vol}(S_j) \leq M.$$

Let $\mathbb{B}^n \subseteq \mathbb{R}^n$ be the closed Euclidean unit ball. Choose an affine isomorphism $\mathcal{T}_j(x) := L_j x + b_j$, with $L_j \in \mathrm{GL}_n(\mathbb{R})$ and $b_j \in \mathbb{R}^n$, which sends the John ellipsoid of S_j to $\mathbb{B}^n$. Writing $\kappa_n := \operatorname{vol}(\mathbb{B}^n)$, John's theorem gives

$$\mathbb{B}^n \subseteq \mathcal{T}_j(S_j) \subseteq n\mathbb{B}^n, \qquad \frac{\kappa_n}{M} \leq |\det L_j| \leq \frac{n^n \kappa_n}{v}.$$

Thus the transformed bodies lie in one fixed compact set, the transformed $A_{i,j}$ have volume bounded away from zero, and their transformed volume gaps tend to zero.

Starting with any subsequence, Blaschke's selection theorem gives a further subsequence along which all $4n$ transformed bodies converge. At the limit,

$$\widehat{A}_i \subseteq \widehat{C}_i, \widehat{D}_i \subseteq \widehat{B}_i, \qquad \operatorname{vol}(\widehat{A}_i) = \operatorname{vol}(\widehat{B}_i) > 0,$$

where the hats denote the respective Hausdorff limits. Hence $\widehat{A}_i = \widehat{C}_i = \widehat{D}_i = \widehat{B}_i$. Continuity of mixed volume now makes the transformed mixed-volume difference tend to zero. Finally,

$$\operatorname{vol}\Big(\mathcal{T}_j(K_1), \ldots, \mathcal{T}_j(K_n)\Big) = |\det L_j| \operatorname{vol}(K_1, \ldots, K_n)$$

for arbitrary $K_1, \ldots, K_n \in \mathcal{K}_n$; the affine translations are harmless. The lower bound for $|\det L_j|$ gives the same conclusion before transformation. Since this works for every subsequence, the full sequence converges. $\square$

The next result upgrades fixed-valuation continuity only at the level of mixed-volume values. It does not assert valuation-uniform Hausdorff convergence of the bodies themselves.

Proposition 4.2. *For $1 \leq i \leq n$, fix a big line bundle L_i, a smooth reference metric with curvature θ_i, and potentials $\varphi_{i,j}, \varphi_i \in \mathrm{PSH}(X, \theta_i)_{>0}$ such that*

$$\varphi_{i,j} \xrightarrow{d_S} \varphi_i.$$

Then

$$\sup_\nu \left| \operatorname{vol}\Big(\Delta_\nu(\theta_1, \varphi_{1,j}), \ldots, \Delta_\nu(\theta_n, \varphi_{n,j})\Big) - \operatorname{vol}\Big(\Delta_\nu(\theta_1, \varphi_1), \ldots, \Delta_\nu(\theta_n, \varphi_n)\Big) \right| \longrightarrow 0, \tag{4.1}$$

where the supremum is over all surjective valuations $\mathbb{C}(X)^\times \to \mathbb{Z}^n_{\mathrm{lex}}$ that are trivial on $\mathbb{C}^\times$.

Proof. Replace every $\varphi_{i,j}$ and φ_i by its model envelope. As recalled in Section 2, this changes neither its d_S-class nor its partial Okounkov body.

Suppose that (4.1) fails. Then there are a subsequence, still indexed by j, valuations ν_j, and a constant $\delta > 0$ for which the absolute difference in (4.1), evaluated at ν_j, is at least δ.

Apply Lemma 2.1 to this subsequence. After passing to a further subsequence, there are increasing model potentials $\eta_{i,j}$ and decreasing model potentials $\psi_{i,j}$ such that

$$\eta_{i,j} \leq \varphi_{i,j} \leq \psi_{i,j}, \qquad \eta_{i,j} \xrightarrow{d_S} \varphi_i, \quad \psi_{i,j} \xrightarrow{d_S} \varphi_i. \tag{4.2}$$

For every i, j, put

$$\begin{aligned} A_{i,j} &:= \Delta_{\nu_j}(\theta_i, \eta_{i,j}), \qquad & B_{i,j} &:= \Delta_{\nu_j}(\theta_i, \psi_{i,j}), \\ C_{i,j} &:= \Delta_{\nu_j}(\theta_i, \varphi_{i,j}), \qquad & D_{i,j} &:= \Delta_{\nu_j}(\theta_i, \varphi_i). \end{aligned}$$

By (2.3) and (4.2),

$$A_{i,j} \subseteq C_{i,j}, D_{i,j} \subseteq B_{i,j}.$$

The numerical bounds in Lemma 4.1 do not depend on ν_j. Indeed, (2.2) and the d_S-continuity of the volume [Xia26a, Corollary 7.3.1] give

$$\operatorname{vol}(B_{i,j}) - \operatorname{vol}(A_{i,j}) = \frac{1}{n!}\Big(\operatorname{vol}(\theta_i + \mathrm{dd^c}\psi_{i,j}) - \operatorname{vol}(\theta_i + \mathrm{dd^c}\eta_{i,j})\Big) \longrightarrow 0,$$

$$\mathrm{vol}(A_{i,j}) = \frac{1}{n!}\mathrm{vol}(\theta_i + \mathrm{dd^c}\eta_{i,j}) \longrightarrow \frac{1}{n!}\mathrm{vol}(\theta_i + \mathrm{dd^c}\varphi_i) > 0.$$

Furthermore, (2.4) and (2.3) yield

$$B_{1,j} + \cdots + B_{n,j} \subseteq \Delta_{\nu_j}\left(\sum_i \theta_i + \mathrm{dd^c}\psi_{i,j}\right) \subseteq \Delta_{\nu_j}(L_1 \otimes \cdots \otimes L_n).$$

By (2.5), the last body has Euclidean volume $\mathrm{vol}(L_1 \otimes \cdots \otimes L_n)/n!$, independently of j and ν_j.

All assumptions of Lemma 4.1 are therefore satisfied. It forces the mixed-volume difference at ν_j to tend to zero, contradicting its lower bound δ. $\square$

5. Proof of the mixed-volume formula

Proposition 5.1. *The conclusion of Theorem 1.2 holds if each h_i has analytic singularities and every $c_1(L_i, h_i)$ is a Kähler current.*

Proof. Take a projective log resolution $\pi\colon Y \to X$, with Y smooth and projective, for which

$$\pi^* c_1(L_i, h_i) = [D_i] + R_i,$$

where D_i is an effective $\mathbb{Q}$-divisor and R_i is a closed positive current with locally bounded potentials; see [Xia26a, Definition 1.6.2 and Theorem 1.6.1]. Put $\alpha_i \coloneqq \{R_i\} \in \mathrm{H}^{1,1}(Y, \mathbb{R})$. The class α_i is nef by [Xia26a, Lemma 1.6.1]. Then

$$\int_X c_1(L_1, h_1) \wedge \cdots \wedge c_1(L_n, h_n) = \int_Y R_1 \wedge \cdots \wedge R_n = \alpha_1 \cap \cdots \cap \alpha_n. \tag{5.1}$$

Here we used [Xia26a, Proposition 2.4.2 and Proposition 2.2.1(1), (3), and (4)]. Note that

$$\alpha_i^n = \int_X c_1(L_i, h_i)^n > 0,$$

Thus $\alpha_i^n > 0$, so the nef class α_i is big.

For a $\mathbb{Q}$-Cartier divisor D and a valuation ν centered at $y_\nu \in Y$, choose $m > 0$ such that mD is Cartier and let g be a local equation of mD at y_ν. Set

$$\nu(D) \coloneqq m^{-1}\nu(g) \in \mathbb{Q}^n.$$

For every valuation ν, the log-resolution formula [Xia26a, Remark 10.3.1, Equation (10.22)] reads

$$\Delta_\nu(L_i, h_i) = \Delta_\nu(\alpha_i) + \nu(D_i). \tag{5.2}$$

Translation invariance, Proposition 3.2, and nefness give

$$n!\,\mathrm{vol}\Big(\Delta_\nu(L_1, h_1), \ldots, \Delta_\nu(L_n, h_n)\Big) \le \langle \alpha_1, \ldots, \alpha_n \rangle = \alpha_1 \cap \cdots \cap \alpha_n. \tag{5.3}$$

It remains to prove the opposite inequality after taking the supremum. Fix an ample $\mathbb{Q}$-divisor H on Y. Choose one $c \in \mathbb{Q}_{>0}$ so large that every $c\alpha_i - \{H\}$ is big, and choose effective $\mathbb{Q}$-divisors E_i in the cohomology class $c\alpha_i - \{H\}$. For $\epsilon \in \mathbb{Q}_{>0}$ set

$$\beta_i \coloneqq \alpha_i + \epsilon\{H\}.$$

The class β_i is ample and

$$\frac{1}{1 + c\epsilon}\beta_i + \frac{\epsilon}{1 + c\epsilon}\{E_i\} = \alpha_i. \tag{5.4}$$

By [Wil25, Theorem 1.1], there is an admissible flag

$$Y_\bullet\colon Y = Y_0 \supset Y_1 \supset \cdots \supset Y_n = \{y\},$$

where each Y_r is irreducible of codimension r in Y and is nonsingular at y. Let ν_ϵ be the valuation defined by this flag. It satisfies

$$n!\,\mathrm{vol}\Big(\Delta_{\nu_\epsilon}(\beta_1), \ldots, \Delta_{\nu_\epsilon}(\beta_n)\Big) = \beta_1 \cap \cdots \cap \beta_n. \tag{5.5}$$

The flag valuation is surjective and trivial on $\mathbb{C}^\times$; since $\mathbb{C}(Y)=\mathbb{C}(X)$, it is among the valuations in Theorem 1.2. Then by (5.4), we have

$$\frac{1}{1+c\epsilon}\Delta_{\nu_\epsilon}(\beta_i)+\frac{\epsilon}{1+c\epsilon}\nu_\epsilon(E_i)\subseteq\Delta_{\nu_\epsilon}(\alpha_i).$$

Using (5.5), we find

$$\sup_\nu \operatorname{vol}\Big(\Delta_\nu(\alpha_1),\dots,\Delta_\nu(\alpha_n)\Big)\geq\frac{1}{n!(1+c\epsilon)^n}\beta_1\cdots\beta_n. \tag{5.6}$$

Letting $\epsilon\to 0+$ in (5.6), and then using the translations in (5.2), proves the reverse of (5.3). Together with (5.1), this proves the proposition. □

Proof of Theorem 1.2. Choose smooth Hermitian metrics h_i^0 on L_i and write

$$\theta_i\coloneqq c_1\left(L_i,h_i^0\right),\qquad h_i=h_i^0\mathrm{e}^{-\varphi_i}.$$

We may replace φ_i by $P_{\theta_i}[\varphi_i]_{\mathcal{I}}$ and assume that φ_i is $\mathcal{I}$-good. [Xia26a, Theorem 7.1.1] gives potentials $\psi_{i,j}$ with analytic singularities such that

$$\psi_{i,j}\xrightarrow{d_S}\varphi_i,\qquad \theta_{i,\psi_{i,j}}\text{ is a Kähler current.}$$

By Proposition 5.1, for every j,

$$\int_X\theta_{1,\psi_{1,j}}\wedge\cdots\wedge\theta_{n,\psi_{n,j}}=n!\sup_\nu\operatorname{vol}\Big(\Delta_\nu(\theta_1,\psi_{1,j}),\dots,\Delta_\nu(\theta_n,\psi_{n,j})\Big). \tag{5.7}$$

By [Xia26a, Theorem 6.2.1], the left-hand side converges to

$$\int_X c_1(L_1,h_1)\wedge\cdots\wedge c_1(L_n,h_n).$$

By Proposition 4.2, the right-hand side converges to

$$n!\sup_\nu\operatorname{vol}\Big(\Delta_\nu(L_1,h_1),\dots,\Delta_\nu(L_n,h_n)\Big).$$

Taking limits in (5.7) proves (1.4). □

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